COMPUTER PROGRAMMING LAB 1

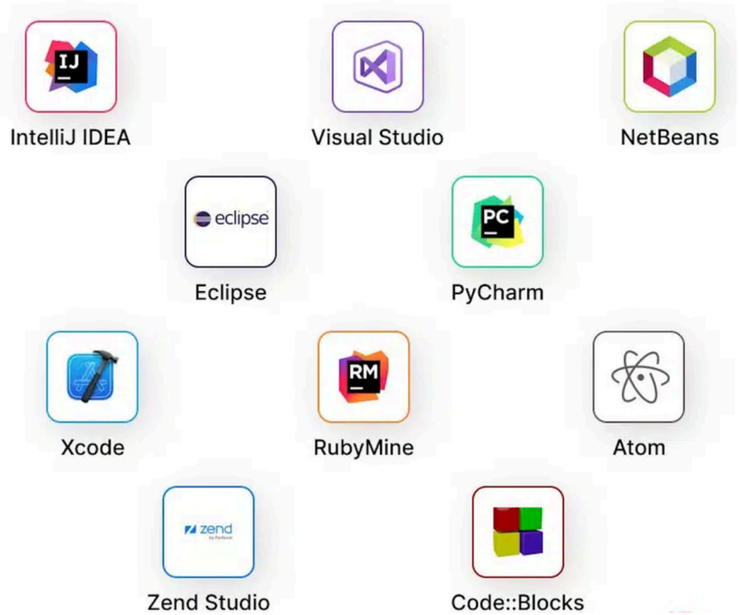
New Uzbekistan University

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What is IDE?

IDE is a workspace that combines programming tools.

- Code editor
- Project explorer
- Compiler integration
- Build tools
- Debugger
- Terminal
- Git integration

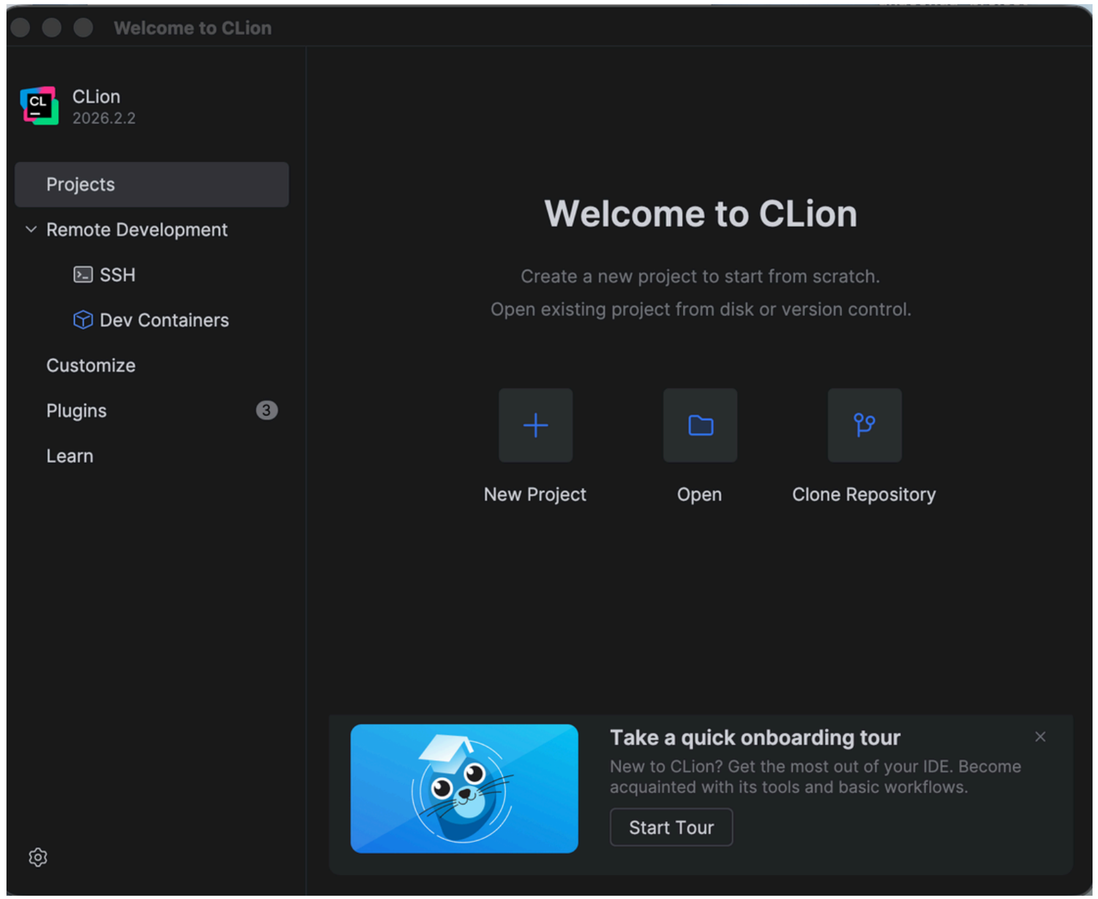


Online compilers

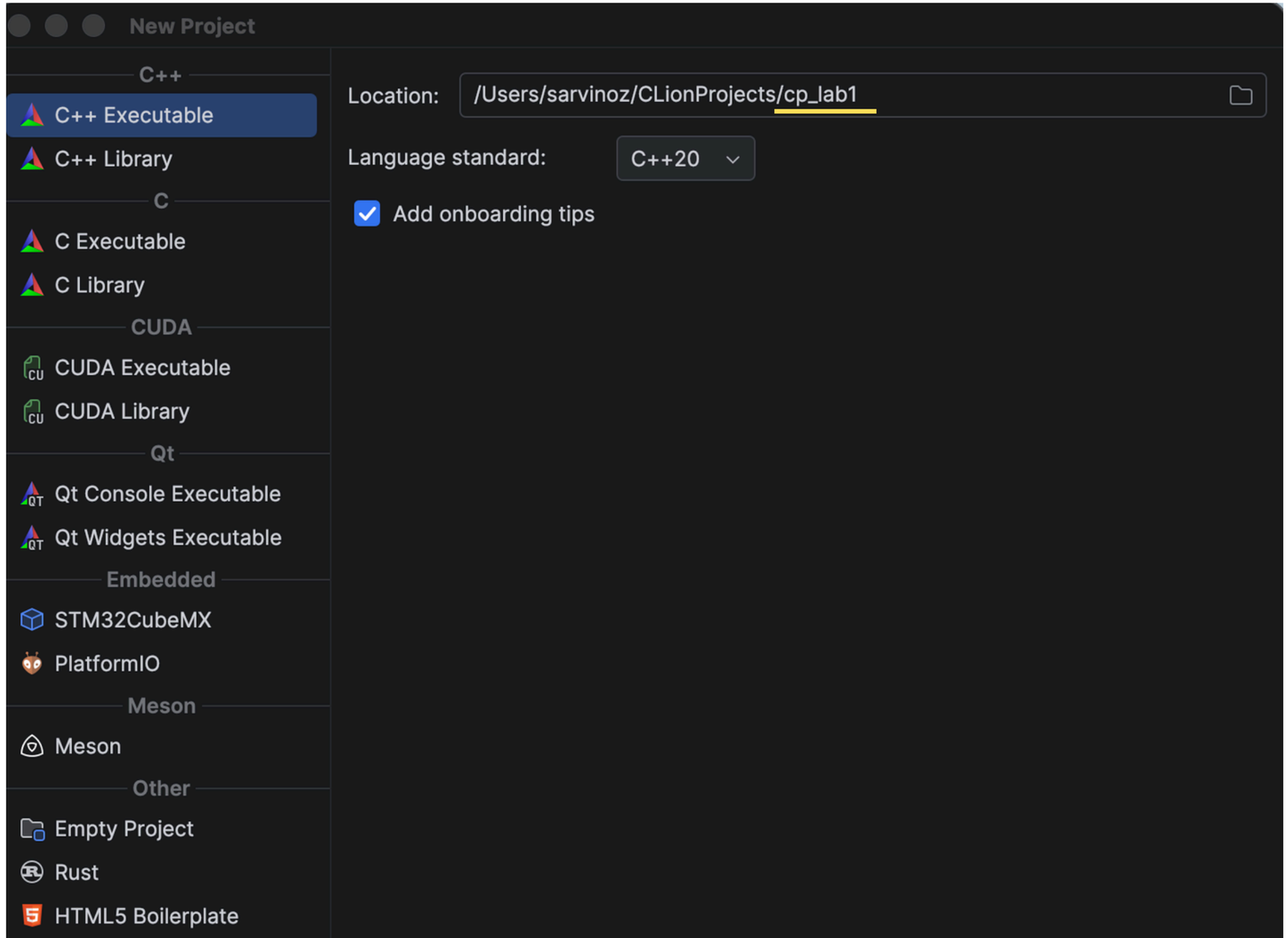
- *Programiz Online C++ Compiler* — simplest interface for a first program.
- *OnlineGDB* — supports compiling, running, multiple files, standard input, and debugging.
- *Compiler Explorer* — useful later for seeing how compilers translate code, but it can look complicated to complete beginners.

Installing CLion

Use the link: https://www.jetbrains.com/clion/download/?section=mac

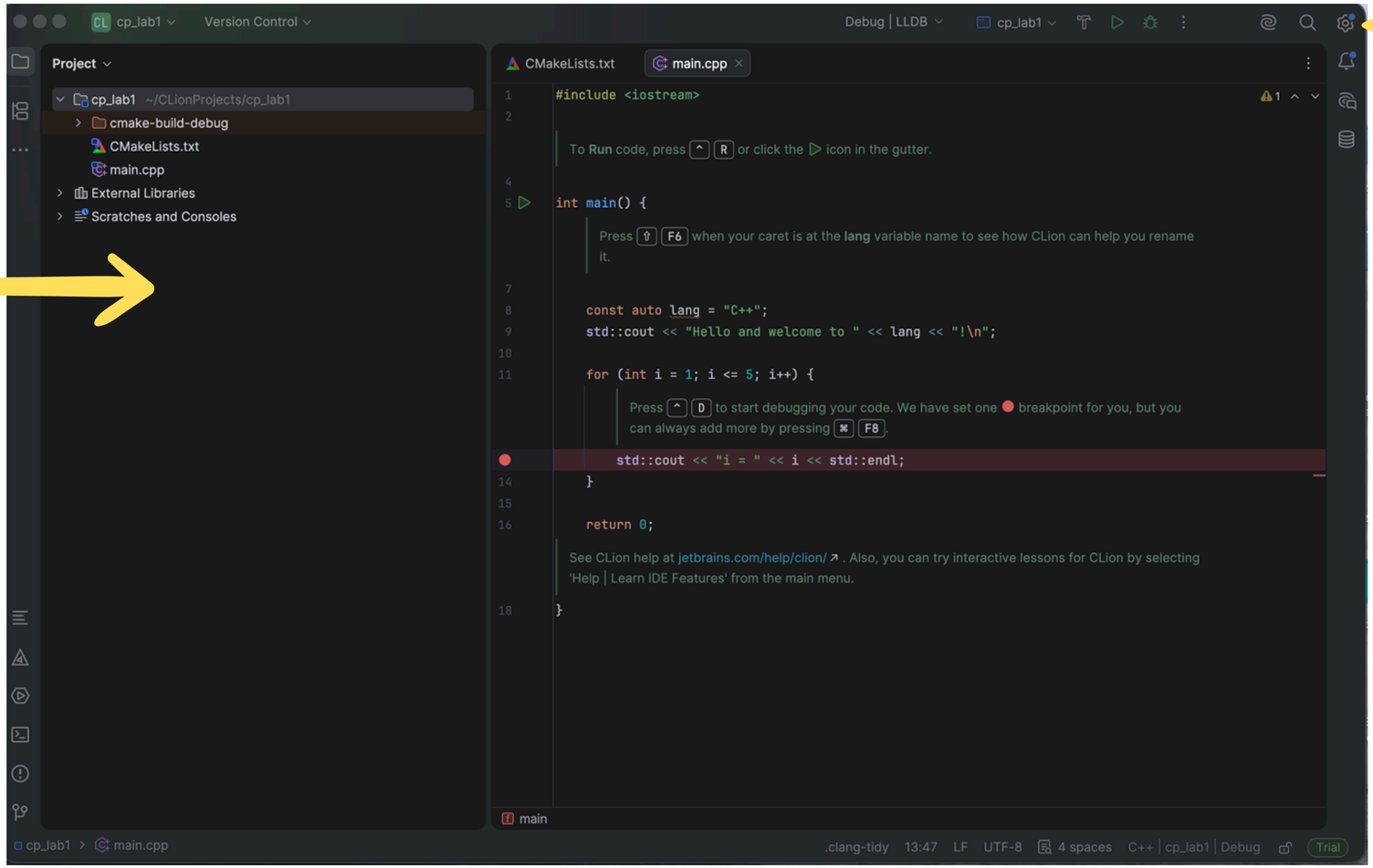


Naming your project



Interface Overview

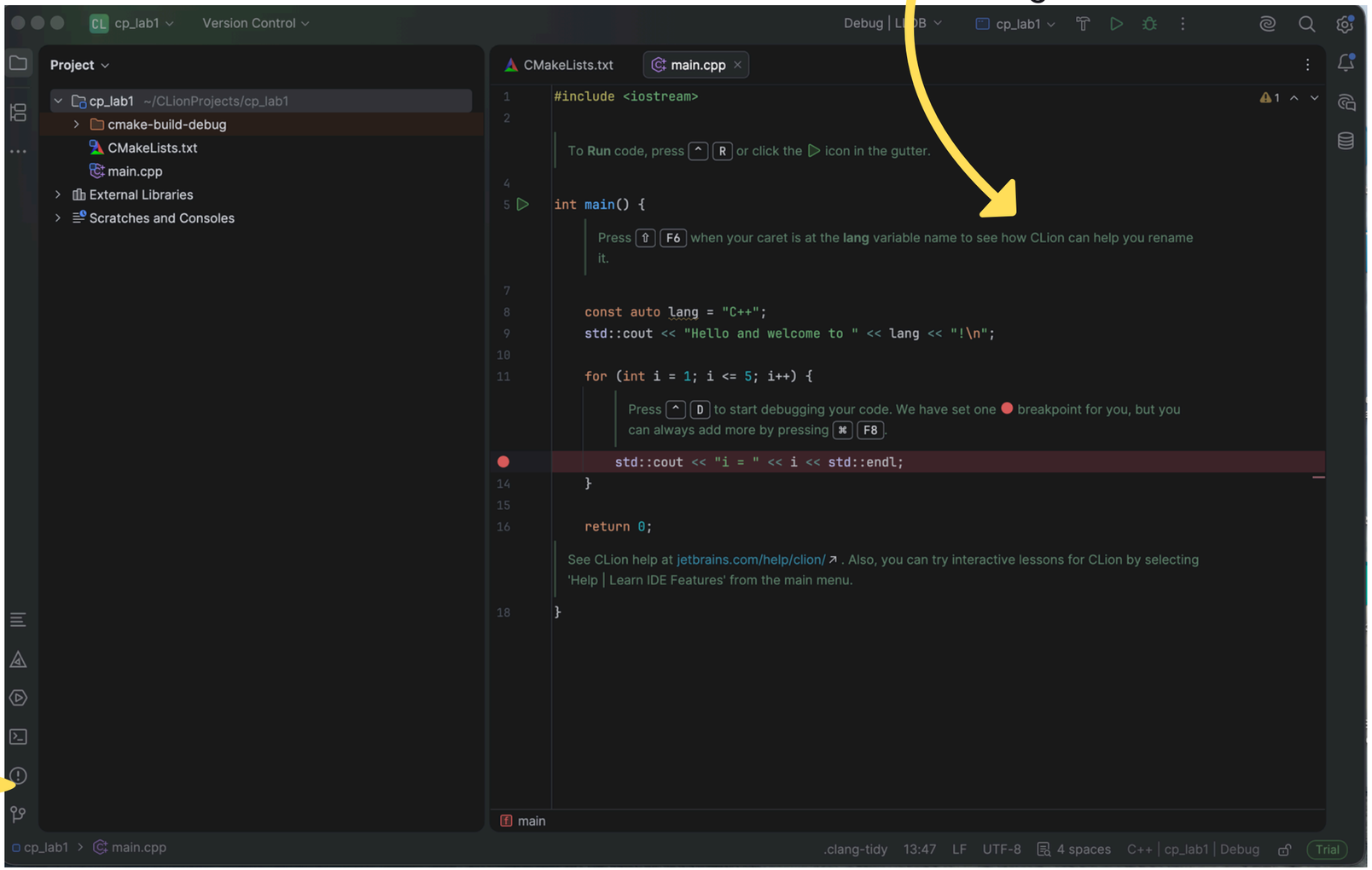
The Project tool window shows the folders and files that belong to our project. It works like a file explorer specifically for the current programming project.



The main toolbar contains the actions we use to build and run our program.

Interface Overview

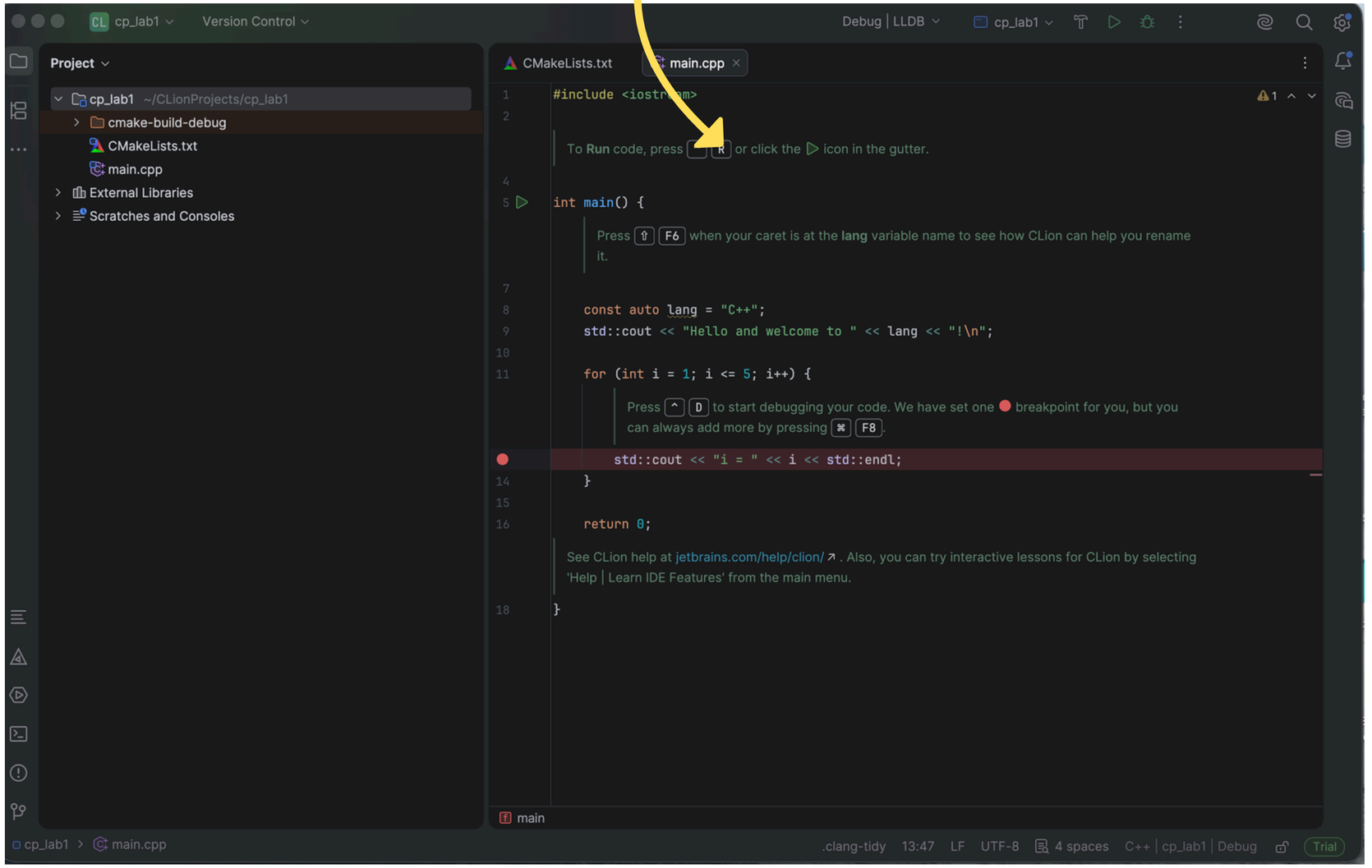
The editor is where we read and write source code. CLion provides syntax highlighting, completion suggestions, error detection, and navigation.



These icons open additional tools, including the terminal, version control, problems, and other project services. We will open them only when needed.

Interface Overview

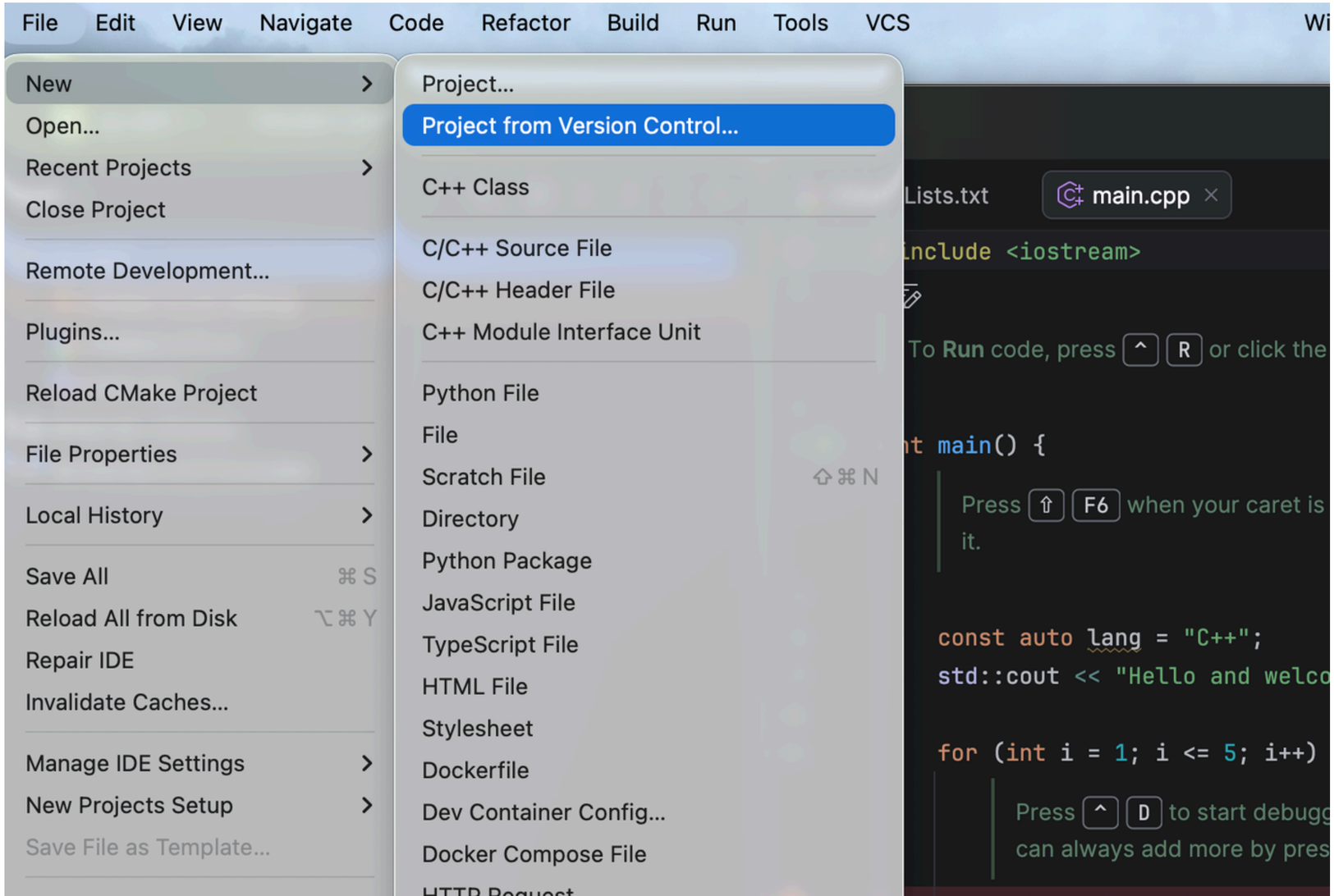
main.cpp is a C++ source file. The .cpp extension tells us that it contains C++ source code. We will write our first program in this file.



These icons open additional tools, including the terminal, version control, problems, and other project services. We will open them only when needed.

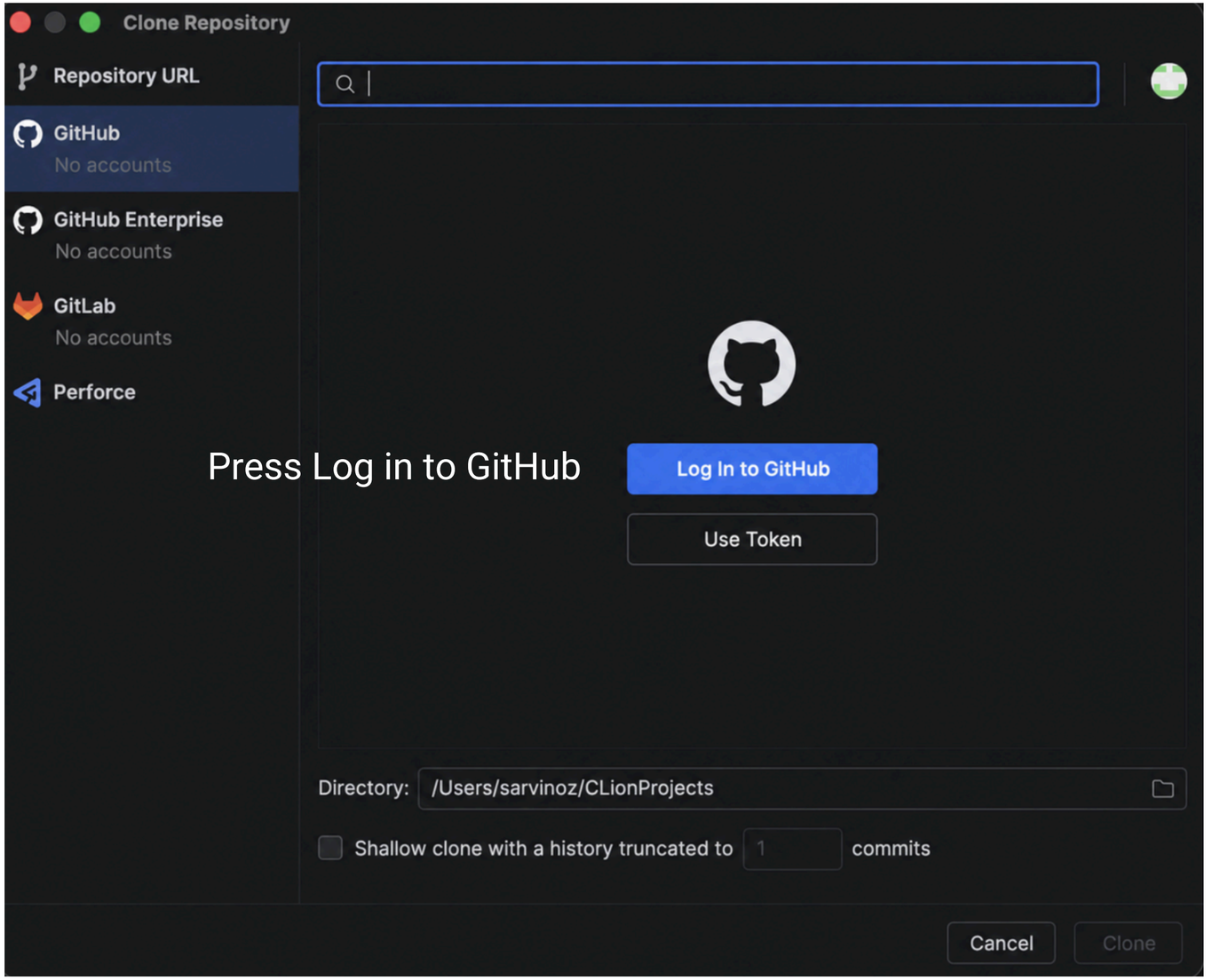
Connecting CLion with GitHub

Go to: File → New → Project from Version Control



Connecting CLion with GitHub

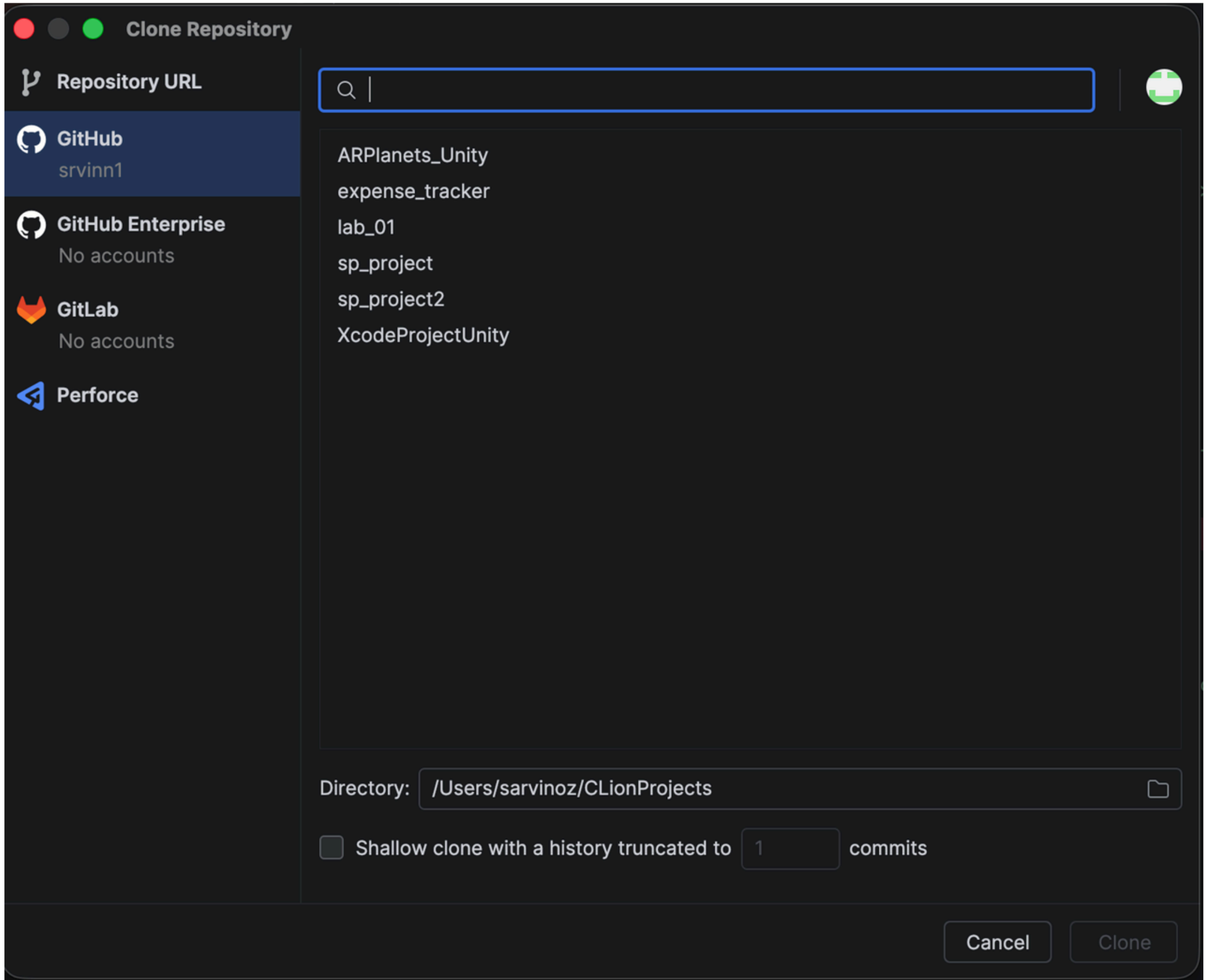
In the left menu select GitHub



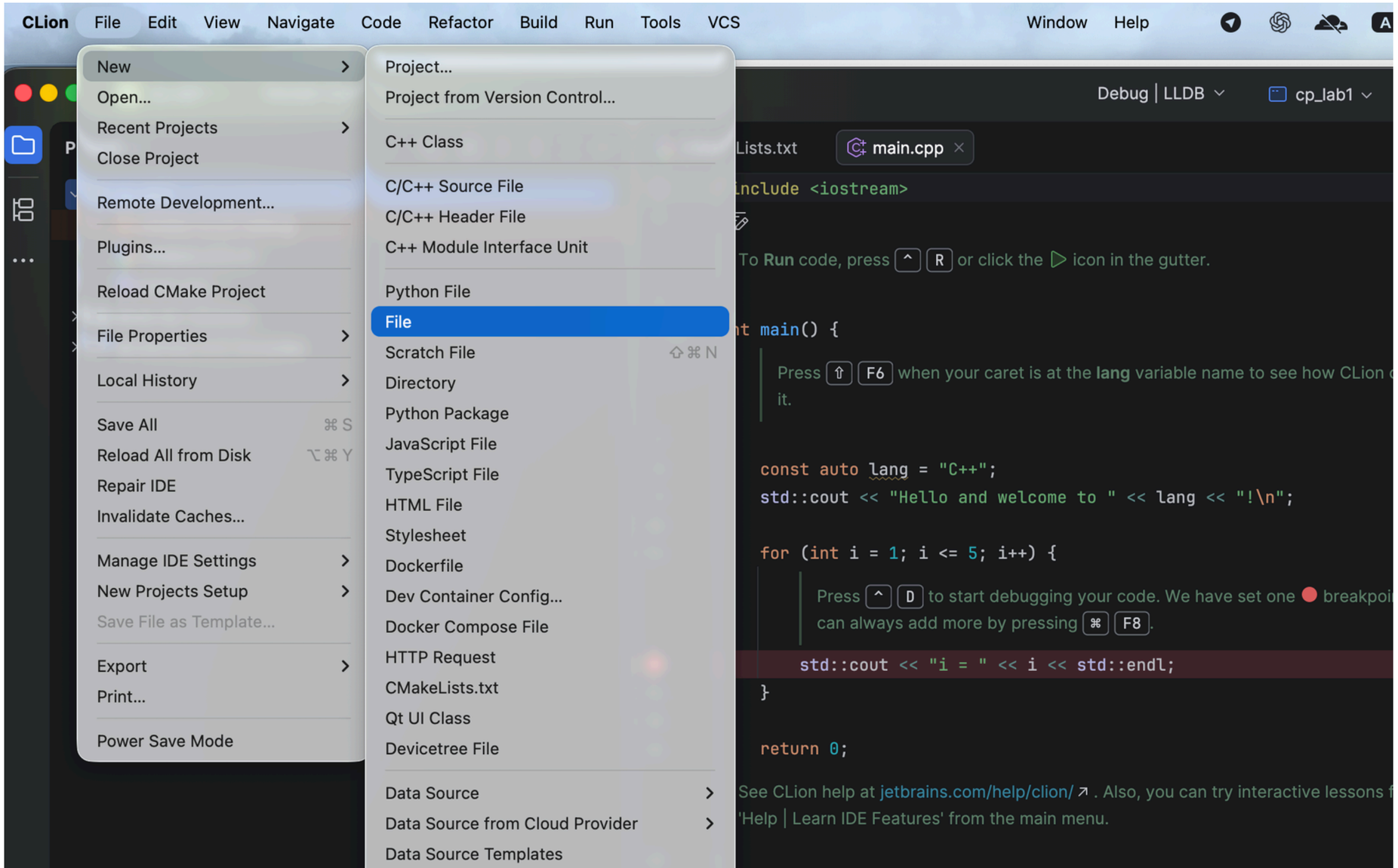
Press Log in to GitHub

Connecting CLion with GitHub

You should see this after logging in.
DO NOT select any repository yet.
We will create it together. Just click “Cancel”

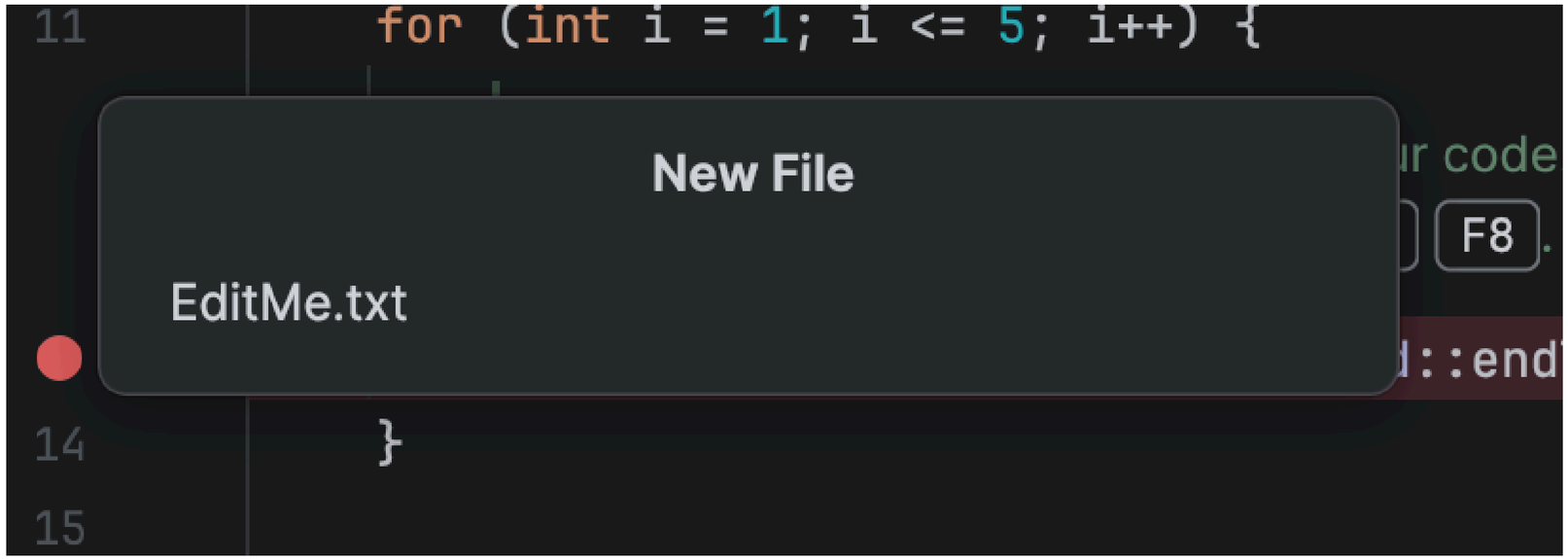


Create a file in Clion



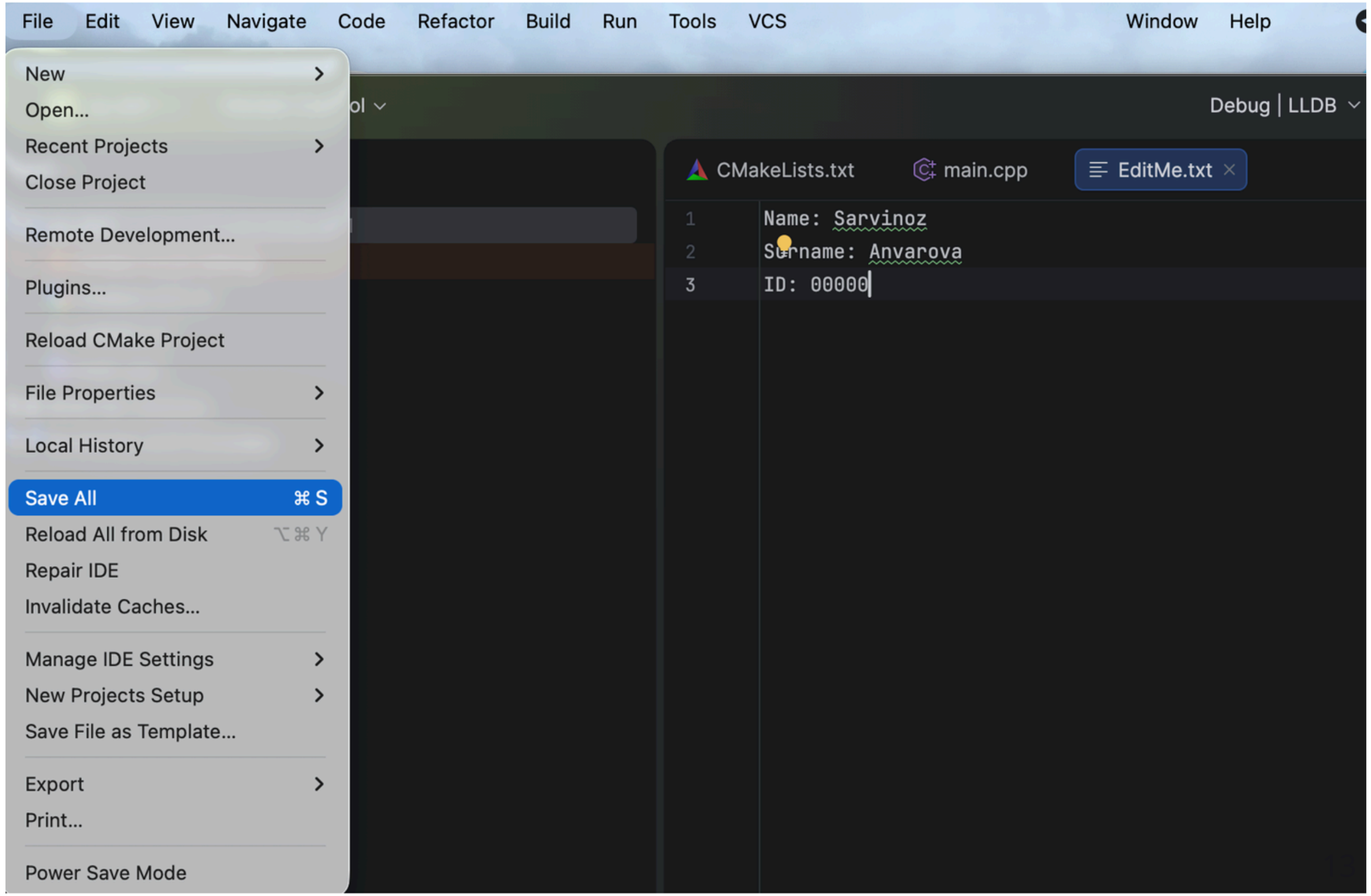
Go to:
File → New → File

Name your file



Saving the file

Write your name, surname and ID inside EditMe.txt file. Save the file using the shortcut command + S or got to File → Save All

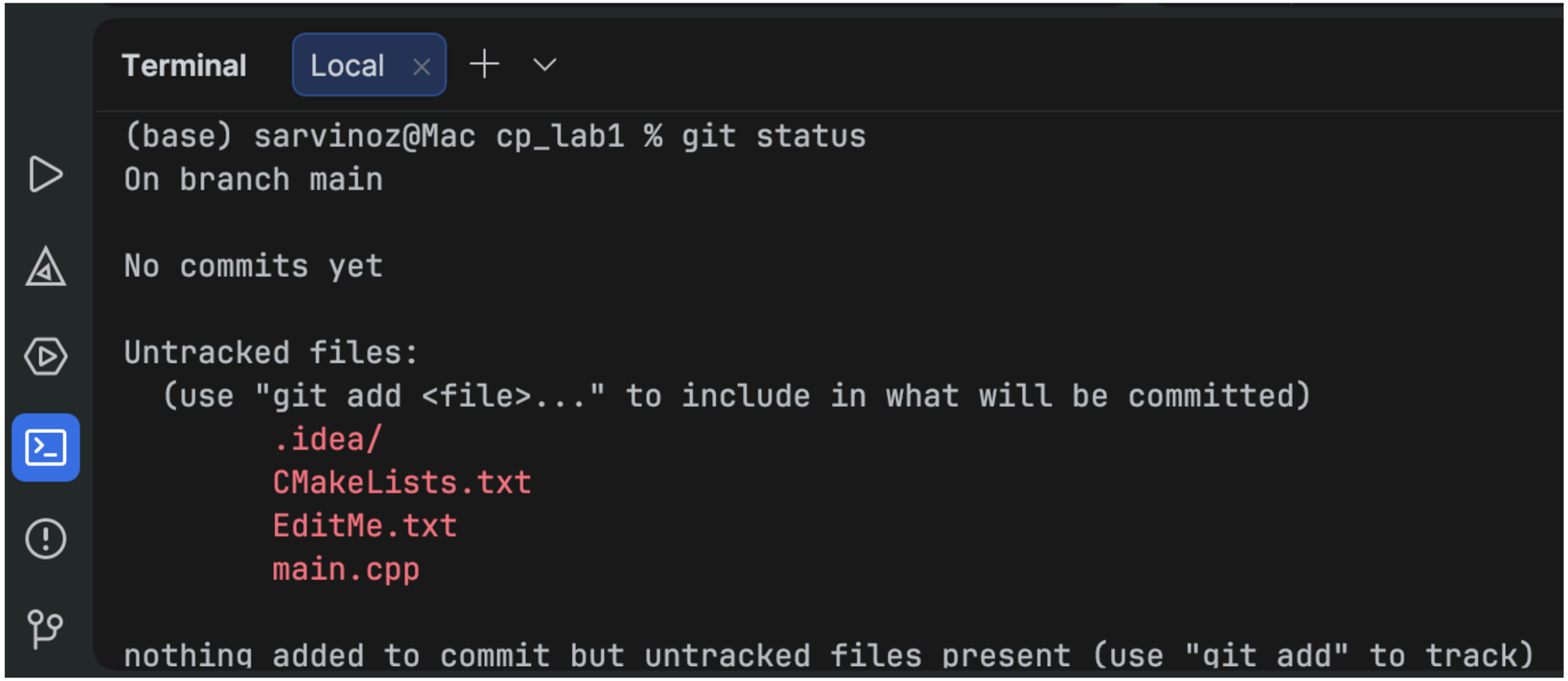


Initialise git

Open terminal in Clion and write **git init** command. This will initialize your Git repository

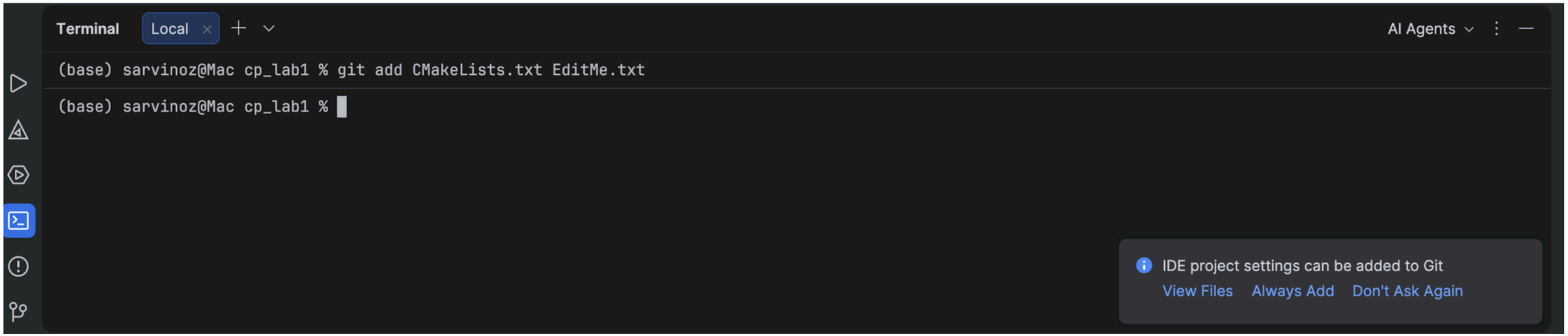


Checking the status of git



This is our status currently we didn’t add files to git and didn’t make a commit yet

Adding files to git



Add these files to the git. No need to add main.cpp yet. We will add it later

Checking the status after git add

```
No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   CMakeLists.txt
        new file:   EditMe.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .idea/
        main.cpp
```

This verifies that we have added our files!

Make your first commit

```
(base) sarvinoz@Mac cp_lab1 % git commit -m "First commit"
[main (root-commit) 5a13ffc] First commit
 2 files changed, 9 insertions(+)
 create mode 100644 CMakeLists.txt
 create mode 100644 EditMe.txt
```

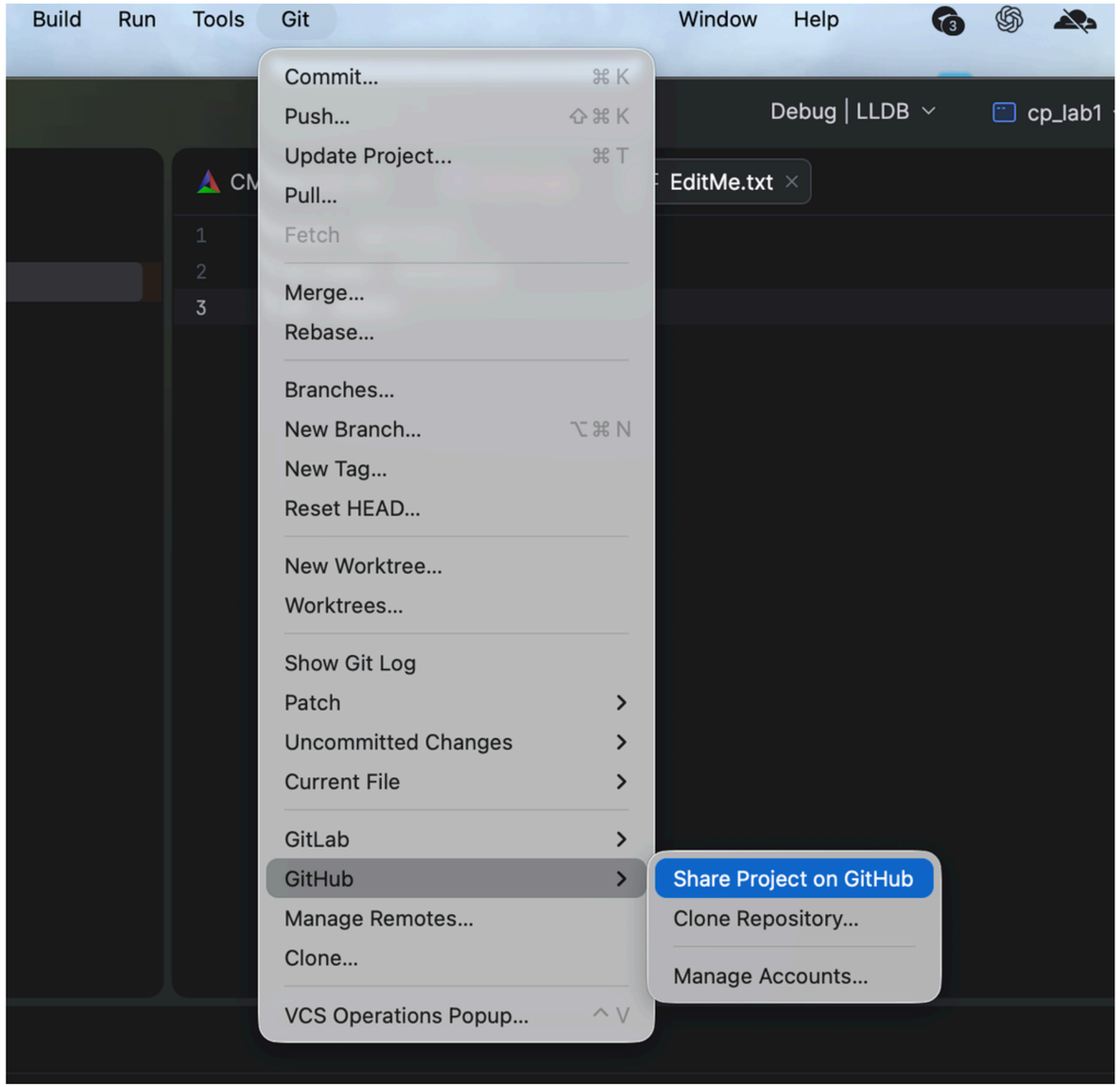
-m is used to define the message. You can write any message inside quotation marks

Verifying commit



Use **git log --oneline** command to verify your commit. If commit is successful you should receive this output which means you are currently on the main branch, and its latest saved version is commit 5a13ffc, named “First commit.”

Sharing it to GitHub



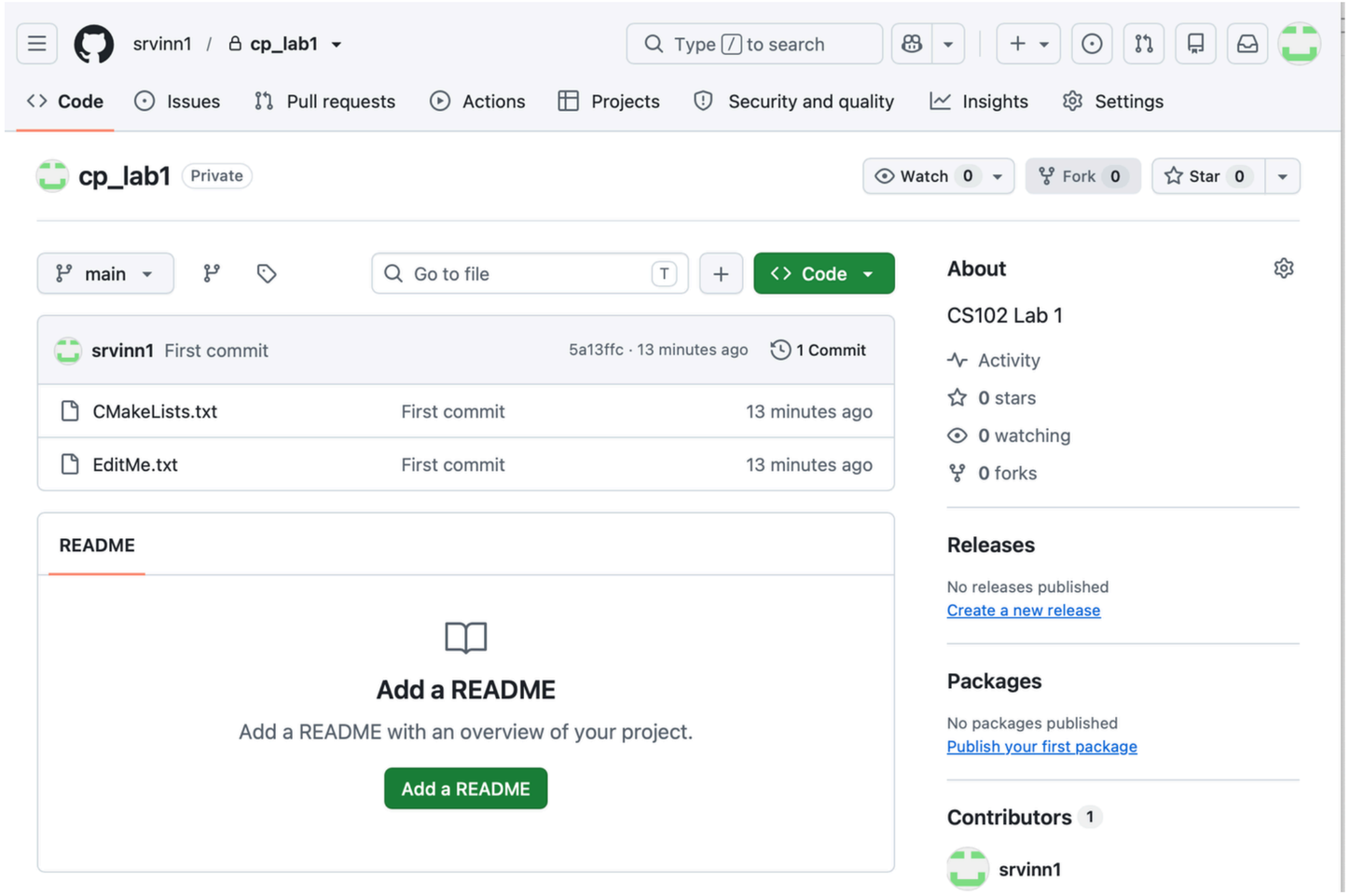
After committing go to Git → GitHub → Share Project to GitHub

Verifying push to GitHub

```
(base) sarvinoz@Mac cp_lab1 % git remote -v
origin  https://github.com/srvinn1/cp_lab1.git (fetch)
origin  https://github.com/srvinn1/cp_lab1.git (push)
```

This message verifies that our push was succesful. You can follow the link and actually see your commit.

Verifying push to GitHub

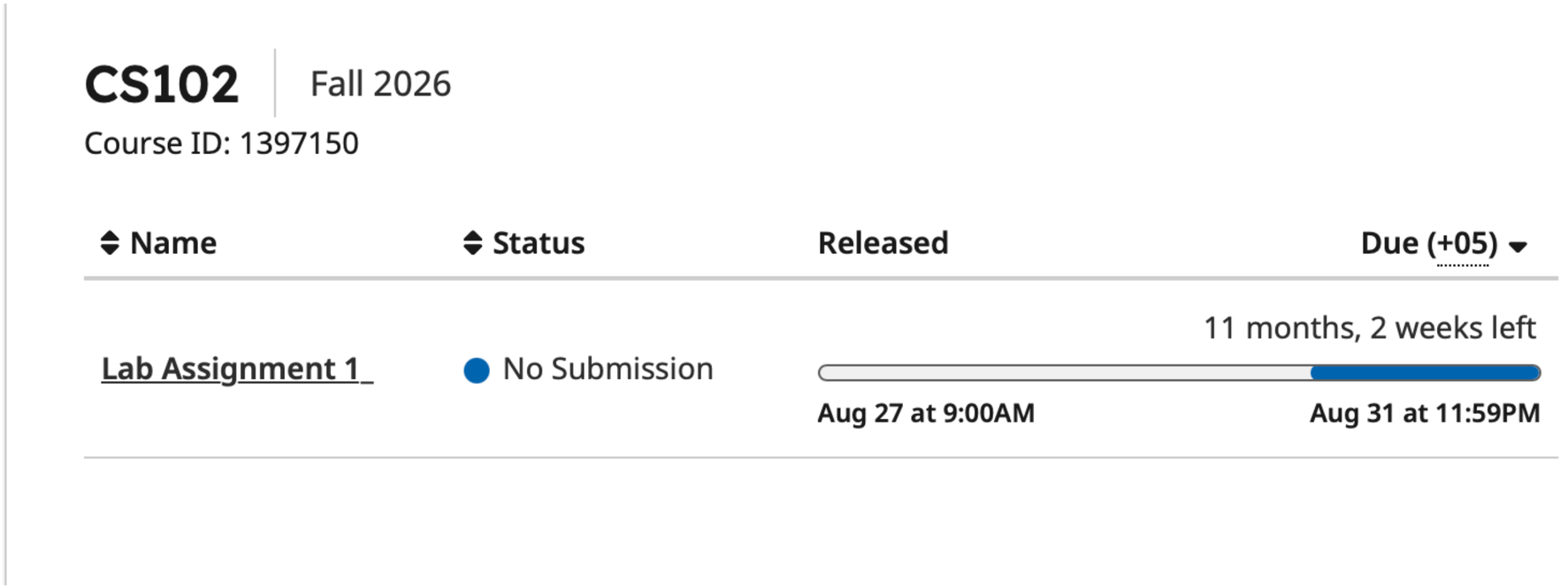


This you should see in your GitHub account. We have created cp_lab1 repository and uploaded files to it

Connecting Gradescope to GitHub

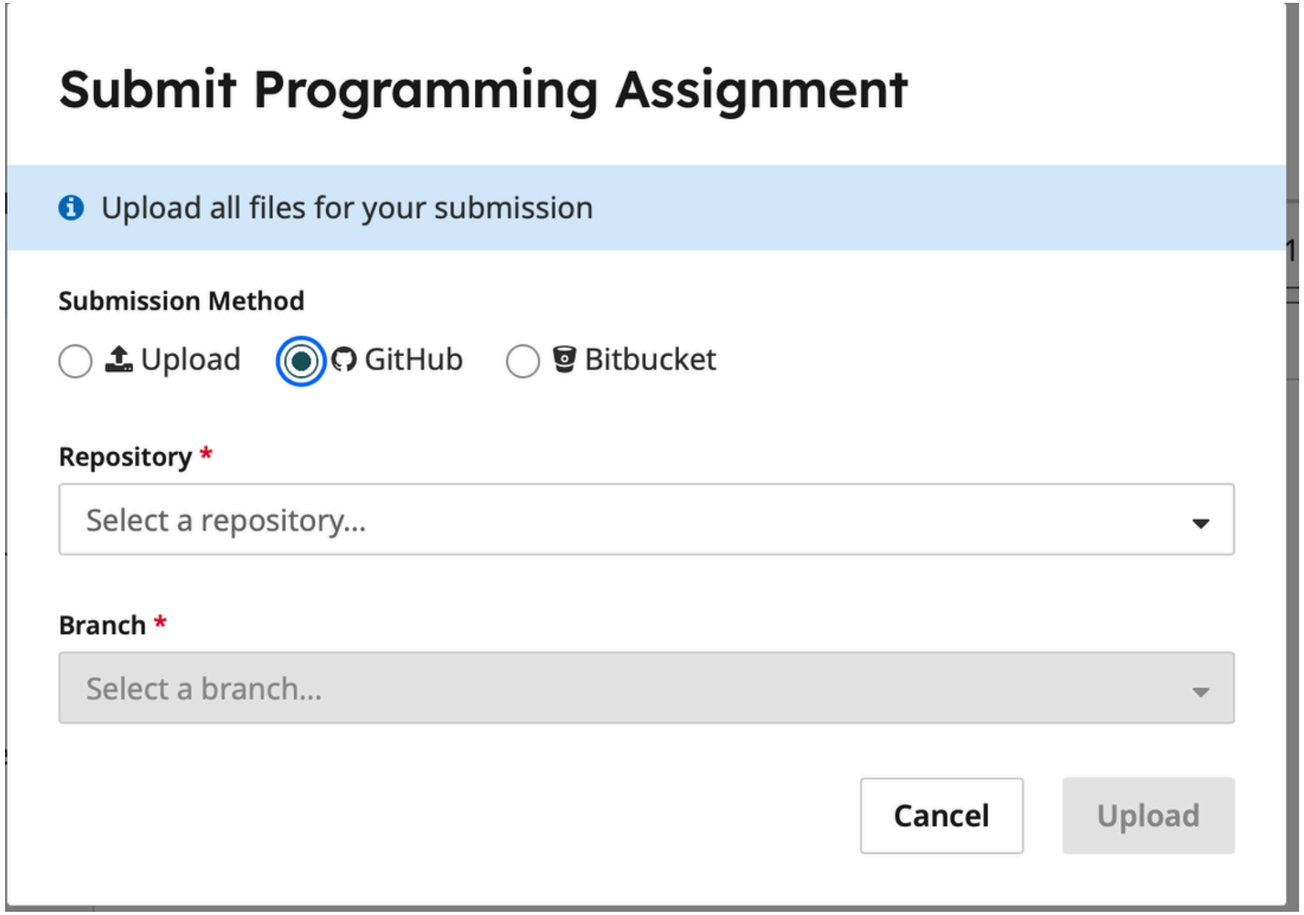
Course Code: RRKJZ8

Link for gradescope: gradescope.com



You should see Fall 2026 course and Lab assignment 1

Connecting Gradescope to GitHub



Click on Lab Assignment 1 and select GitHub as submission method. You will be redirected to authorizing page. Authorize but do not submit anything yet

Thank you!